```
import numpy as np
import matplotlib.pyplot as plt

from sklearn.datasets import load_breast_cancer
from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier, plot_tree
from sklearn.metrics import accuracy_score

# Load dataset
data = load_breast_cancer()

X = data.data
y = data.target

print("Dataset shape:", X.shape)
```

```
Dataset shape: (569, 30)
```

```
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

depths = range(1, 16)

train_accuracy = []
test_accuracy = []

for depth in depths:
    model = DecisionTreeClassifier(max_depth=depth, random_state=42)
    model.fit(X_train, y_train)

    # Predictions
    y_train_pred = model.predict(X_train)
    y_test_pred = model.predict(X_test)

    # Accuracy
    train_accuracy.append(accuracy_score(y_train, y_train_pred))
    test_accuracy.append(accuracy_score(y_test, y_test_pred))

for depth, train_acc, test_acc in zip(depths, train_accuracy, test_accuracy):
    print(f"Depth {depth}: Training Accuracy = {train_acc:.4f}, Testing
Accuracy = {test_acc:.4f}")
```

```
Depth 1: Training Accuracy = 0.9209, Testing Accuracy = 0.8947
Depth 2: Training Accuracy = 0.9297, Testing Accuracy = 0.9298
Depth 3: Training Accuracy = 0.9780, Testing Accuracy = 0.9474
Depth 4: Training Accuracy = 0.9956, Testing Accuracy = 0.9474
Depth 5: Training Accuracy = 0.9956, Testing Accuracy = 0.9474
Depth 6: Training Accuracy = 0.9978, Testing Accuracy = 0.9386
Depth 7: Training Accuracy = 1.0000, Testing Accuracy = 0.9474
Depth 8: Training Accuracy = 1.0000, Testing Accuracy = 0.9474
Depth 9: Training Accuracy = 1.0000, Testing Accuracy = 0.9474
Depth 10: Training Accuracy = 1.0000, Testing Accuracy = 0.9474
Depth 11: Training Accuracy = 1.0000, Testing Accuracy = 0.9474
Depth 12: Training Accuracy = 1.0000, Testing Accuracy = 0.9474
Depth 13: Training Accuracy = 1.0000, Testing Accuracy = 0.9474
Depth 14: Training Accuracy = 1.0000, Testing Accuracy = 0.9474
Depth 15: Training Accuracy = 1.0000, Testing Accuracy = 0.9474
```

```
for i in range(1, len(depths)):
    if test_accuracy[i] < test_accuracy[i-1] and train_accuracy[i] >
train_accuracy[i-1]:
        print("Overfitting starts around tree depth:", depths[i])
        break
```

```
Overfitting starts around tree depth: 6
```

```
best_depth = depths[np.argmax(test_accuracy)]

print("Best tree depth:", best_depth)
print("Best testing accuracy:", max(test_accuracy))
```

```
Best tree depth: 3
Best testing accuracy: 0.9473684210526315
```

```
model = DecisionTreeClassifier(max_depth=best_depth, random_state=42)
model.fit(X_train, y_train)

plt.figure(figsize=(20, 10))

plot_tree(
    model,
    filled=True,
    feature_names=data.feature_names,
    class_names=data.target_names
)

plt.show()
```

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Practical No.-9

